

05 Integration and Interface Specification

ALDRIDGE HARBOR PORT AUTHORITY

KESSLER POINT MARINE TERMINAL — PERIMETER SECURITY, ACCESS CONTROL AND SURVEILLANCE SYSTEMS

DOCUMENT 5 OF 13 — INTEGRATION AND INTERFACE SPECIFICATION

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ABBREVIATIONS AND ACRONYMS

Term	Meaning
AIS	Automatic Identification System
ALPR	Automatic Licence Plate Recognition
API	Application Programming Interface
CODECO	Container Gate-In/Gate-Out Report (UN/EDIFACT)
COPARN	Container Announcement (UN/EDIFACT)
EDI	Electronic Data Interchange
GOS	Gate Operating System
NCIC	National Crime Information Center
PSIM	Physical Security Information Management
REST	Representational State Transfer
TOS	Terminal Operating System
TSA	Transportation Security Administration
TWIC	Transportation Worker Identification Credential
VBS	Vehicle Booking System
VMS	Video Management System
XML	Extensible Markup Language

1. GENERAL

- IF-1.1** This document defines the interfaces between the Works and systems provided by others.
- IF-1.2** The Contractor shall design, develop, test and commission all interfaces described in this document.
- IF-1.3** The Contractor shall be responsible for obtaining from each third-party system provider such interface specifications, licences, test environments and support as it requires.
- IF-1.4** The Authority will facilitate an introduction to each third-party system provider. The Authority gives no warranty as to the willingness, timescale or commercial terms of any such provider.
- IF-1.5** Each interface shall be documented in an Interface Control Document submitted for approval at Critical Design Review.
- IF-1.6** Each interface shall be tested end to end during the Site Acceptance Test.

2. INTERFACE 1 — TERMINAL OPERATING SYSTEM

2.1 Purpose

IF-2.1 The security pre-gate shall query the Terminal Operating System to establish whether the driver and vehicle presenting at the lane have a valid reason to enter the Site.

2.2 Requirements

IF-2.2 The Contractor shall integrate the security system with the Terminal Operating System in use at the Site.

IF-2.3 The interface shall support a synchronous query returning, for a given credential identifier and vehicle identifier: whether a booking or appointment exists; the appointment window; the transaction type; and the expected container, where one is associated.

IF-2.4 The interface shall return a response within **two (2) seconds** of the query being issued.

IF-2.5 The interface shall support not fewer than **fifty (50)** queries per minute.

IF-2.6 The interface shall use a secure transport with mutual authentication.

IF-2.7 The Contractor may propose REST, web services, UN/EDIFACT message exchange (COPARN, CODECO) or XML messaging, subject to the capability of the Terminal Operating System.

IF-2.8 The Contractor shall not require any modification to the Terminal Operating System.

IF-2.9 Where the Terminal Operating System is unavailable, the pre-gate shall continue to operate and shall record the transaction for subsequent reconciliation.

3. INTERFACE 2 — GATE OPERATING SYSTEM

3.1 Purpose

IF-3.1 The Gate Operating System is provided by others and is outside the scope of this Contract. The security system shall provide the Gate Operating System with the identity of the driver admitted at the pre-gate, so that the commercial transaction may proceed without a second credential presentation.

3.2 Requirements

IF-3.2 On admitting a vehicle, the security system shall transmit to the Gate Operating System: the driver identifier; the credential expiry; the vehicle licence plate; the chassis identifier; the RFID tractor tag; the lane and time of admission; and the pre-gate decision reference.

IF-3.3 The transmission shall occur within **one (1) second** of the barrier being raised.

IF-3.4 The interface shall be resilient to the unavailability of the Gate Operating System, queuing messages for delivery on restoration.

IF-3.5 The security system shall not receive, process or store any commercial data from the Gate Operating System.

4. INTERFACE 3 — LAW ENFORCEMENT WATCHLIST

4.1 Purpose

IF-4.1 The security pre-gate shall establish whether the driver or the vehicle presenting at the lane is the subject of a law enforcement record of interest.

4.2 Requirements

IF-4.2 The system shall query the **National Crime Information Center (NCIC)** for each driver presenting at the pre-gate, using the identity data read from the credential.

IF-4.3 The system shall query the National Crime Information Center for each vehicle presenting at the pre-gate, using the licence plate read by the optical character recognition portal.

IF-4.4 A positive response shall be presented to the Command and Control operator and shall be logged.

IF-4.5 The interface shall return a response within **three (3) seconds**.

IF-4.6 **Where the watchlist interface is unavailable, the pre-gate shall fail closed and no vehicle shall be admitted until the interface is restored.**

IF-4.7 The system shall additionally check each credential against the Transportation Security Administration's TWIC Canceled Card List.

IF-4.8 The system shall support a locally maintained deny list, administered by the Facility Security Officer.

IF-4.9 Watchlist query results shall be retained for **twelve (12) months**.

5. INTERFACE 4 — SURFACE RADAR AND AIS

IF-5.1 The Contractor shall integrate the waterside surface radar with the video management system such that a radar track within the defined security zone cues the nearest thermal camera and the nearest PTZ camera to the track position.

IF-5.2 The Contractor shall integrate an Automatic Identification System receiver such that radar tracks correlated with a declared AIS vessel identity are distinguished in the operator display from uncorrelated tracks.

IF-5.3 An uncorrelated track entering the security zone shall raise an alarm.

IF-5.4 The AIS receiver is within the scope of this Contract.

6. INTERFACE 5 — FIRE ALARM SYSTEM

IF-6.1 The Contractor shall interface the access control system to the Site fire alarm system such that, on a fire alarm condition, all turnstiles release to free rotation and all electrically locked doors on designated egress routes unlock.

IF-6.2 The interface shall be by hardwired fail-safe connection. A network interface shall not be used for this function.

IF-6.3 The fire alarm system is provided by others.

7. INTERFACE 6 — AUTHORITY SECURITY INFORMATION AND EVENT MANAGEMENT

IF-7.1 The Contractor shall forward security-relevant logs from all supplied systems to the Authority's Security Information and Event Management platform.

IF-7.2 Logs shall be forwarded in syslog or Common Event Format over a secure transport.

IF-7.3 The Contractor shall agree the log content and format with the Authority's information technology department.

8. INTERFACE 7 — PHYSICAL SECURITY INFORMATION MANAGEMENT

IF-8.1 All supplied subsystems shall be integrated to the Physical Security Information Management platform, presenting alarms, device status, video and access events in a single operator interface.

IF-8.2 The platform shall present each alarm with the associated live video, the relevant standard operating procedure, and a record of the operator's actions.

IF-8.3 Integration shall be by the subsystem manufacturer's published interface. Where no published interface exists, the Contractor shall develop one.

IF-8.4 The platform shall be capable of accepting a further subsystem in future without modification to existing integrations.

9. INTERFACE RESPONSIBILITY MATRIX

Interface	Security system side	Third-party side	Interface specification	Commercial responsibility
1 — Terminal Operating System	Contractor	TOS provider	<i>To be obtained by the Contractor</i>	<i>Not stated</i>
2 — Gate Operating System	Contractor	GOS provider	<i>To be obtained by the Contractor</i>	<i>Not stated</i>
3 — Law enforcement watchlist	Contractor	<i>Not stated</i>	<i>Not stated</i>	<i>Not stated</i>
4 — Radar and AIS	Contractor	Contractor	Contractor	Contractor
5 — Fire alarm	Contractor	Others	Existing	Contractor
6 — SIEM	Contractor	Authority IT	Authority	Authority
7 — PSIM	Contractor	Contractor	Contractor	Contractor

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